

The CRISP-DM Process Model

Project: Real-Time Human Activity Recognition from Smartphone Sensors

HW2 — Data Understanding and Data Preparation

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Data Understanding

2.1 Initial Data Collection Report

We use two complementary data sources in this project. The primary source for training and evaluation is the UCI HAR Dataset (Anguita et al., 2013), a publicly available, peer-reviewed dataset distributed under a non-commercial research license. The secondary source is a set of live recordings we collected with the Phyphox mobile application, which serves as an external live demo of the trained model on a new device and a new user.

Dataset metadata

The table below summarises the formal metadata of the two data sources. All numbers come from our own analysis of the files; the code that produced them is included with the submission.

Attribute	UCI HAR Dataset	Phyphox recordings
Source	UCI Machine Learning Repository (Anguita et al., 2013)	Recorded by our team using the Phyphox app, April 2026
License	Public, research/educational use only; any commercial use prohibited	Team-owned
Subjects	30 volunteers, ages 19–48 (21 train / 9 test, disjoint)	1 (one team member)
Device	Samsung Galaxy S II worn on the waist	Personal smartphone; position varies by recording
Sensors	Triaxial accelerometer + gyroscope	Triaxial accelerometer or gyroscope (one per file)
Sampling rate	50 Hz (constant)	~500 Hz (measured 499.9–500.7 Hz); we resampled to 50 Hz
Units	Accel: g (1 g = 9.80665 m/s ²); Gyro: rad/s	Accel: m/s ² (we converted to g); Gyro: rad/s
Windowing	Fixed: 2.56 s (128 samples), 50% overlap (already done)	We applied: 128 samples @ 50 Hz, 50% overlap
Features	561 engineered features per window, in [-1, 1]	None — raw signal only
Activities	6: walking, walking upstairs, walking downstairs, sitting, standing, laying	4: walking (hand or pocket), sitting, standing, laying
Total records	7,352 train + 2,947 test = 10,299 windows	9 recordings ≈ 18.1 min total; ~800 windows after our pipeline
Missing values	None	None (in raw values); see §2.4 on missing sensors

Role in project	Train + held-out test	External live demo (generalization check)
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UCI HAR Dataset

We downloaded the dataset from the UCI Machine Learning Repository as `human_activity_recognition_using_smartphones.zip`. It was collected from 30 volunteers (aged 19–48) performing six activities while wearing a Samsung Galaxy S II on the waist. The accelerometer and gyroscope were sampled at a constant 50 Hz, and the activities were manually labelled from synchronized video. The authors pre-processed the signals (noise filtering, gravity/body separation, sliding windows of 2.56 s with 50% overlap) and provided a 561-element feature vector for each window. The data ships as plain-text files split into a training set (7,352 windows from 21 subjects) and a test set (2,947 windows from 9 subjects). The 9 raw inertial channels per window are also included, which leaves us the option of working with the engineered features, with the raw windows, or with both.

Our Phyphox recordings

We collected nine recordings on 26 April 2026, using the open-source Phyphox app. Each recording lasts about two minutes and captures either the linear accelerometer (with gravity, m/s^2) or the gyroscope (rad/s) at the phone's native sensor rate (~ 500 Hz). We covered four activities — walking, sitting, standing and lying — and we recorded walking twice: once with the phone in hand and once with the phone in the front pocket, so that we could compare the two phone positions against UCI's waist-mounted setup. Each file is a legacy .xls workbook with a Raw Data sheet (Time, x, y, z, absolute magnitude) and two metadata sheets (device information and absolute timestamps).

Attribute importance

We treat all accelerometer and gyroscope axes as equally important at this stage; we will revisit per-axis informativeness during feature analysis. Subject ID is highly important to us because train/test must be split by subject — never by window — to avoid leakage. Activity label is the target. We drop the absolute-magnitude column in Phyphox because it is the Euclidean norm of x, y, z and we can recompute it if we need it.

2.2 Data Description Report

UCI HAR Dataset — gross properties

- Format: plain text, space-separated values, one window per row.
- Training set X_train.txt: 7,352 rows × 561 columns (engineered features).
- Test set X_test.txt: 2,947 rows × 561 columns.
- Labels (y_train.txt / y_test.txt): integers 1–6 mapped to WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING.
- Subject IDs (subject_train.txt / subject_test.txt): integers 1–30. Training: 21 unique IDs; test: 9 unique IDs; no overlap (we verified this).
- Raw inertial signals (9 channels: body_acc {x,y,z}, body_gyro {x,y,z}, total_acc {x,y,z}): each file is $7,352 \times 128$ for train and $2,947 \times 128$ for test, i.e. 128 samples per window at 50 Hz = 2.56 s.
- Features are pre-normalised to [-1, 1]; raw inertial values are in 'g' (accelerometer) and rad/s (gyroscope).
- All 561 feature names are listed in features.txt. We found that 42 names appear more than once (84 columns share names with other columns) — these are the bandsEnergy() features computed on different frequency bands. We rename duplicates at load time so the columns can be addressed unambiguously in pandas.

Our Phyphox recordings — gross properties

- Format: legacy Excel .xls (BIFF), one workbook per recording, three sheets each: Raw Data, Metadata Device, Metadata Time.
- Nine recordings: four accelerometer files (sitting, standing, lying, walking-in-hand) and five gyroscope files (sitting, standing, lying, walking-in-hand, walking-in-pocket).
- Per-file record count ranges from 58,952 to 63,827 rows; durations from 117.8 s to 127.5 s; total $\approx$ 18.1 minutes.
- Measured sampling rate (we computed it from the Time column): 499.9–500.7 Hz across all files, average $\approx$ 500.4 Hz.
- Raw Data fields: Time (s); per-axis values; absolute magnitude.
- Units: accelerometer in m/s^2 ; gyroscope in rad/s. We convert accelerometer values to g (divide by 9.80665) so the unit matches UCI.

2.3 Data Exploration Report

We explored the UCI training set first and confirmed that the six classes are reasonably balanced. The largest class (LAYING) has 1,407 windows and the smallest (WALKING_DOWNSTAIRS) has 986, giving a max/min ratio of 1.43 — well within the range where stratified metrics remain meaningful without resampling. The per-subject window count in the training set ranges from 281 to 409 with a mean of 350.1, so no single subject dominates. Figure 1 shows the distribution; the bar shapes confirm that the train/test split preserves class proportions.

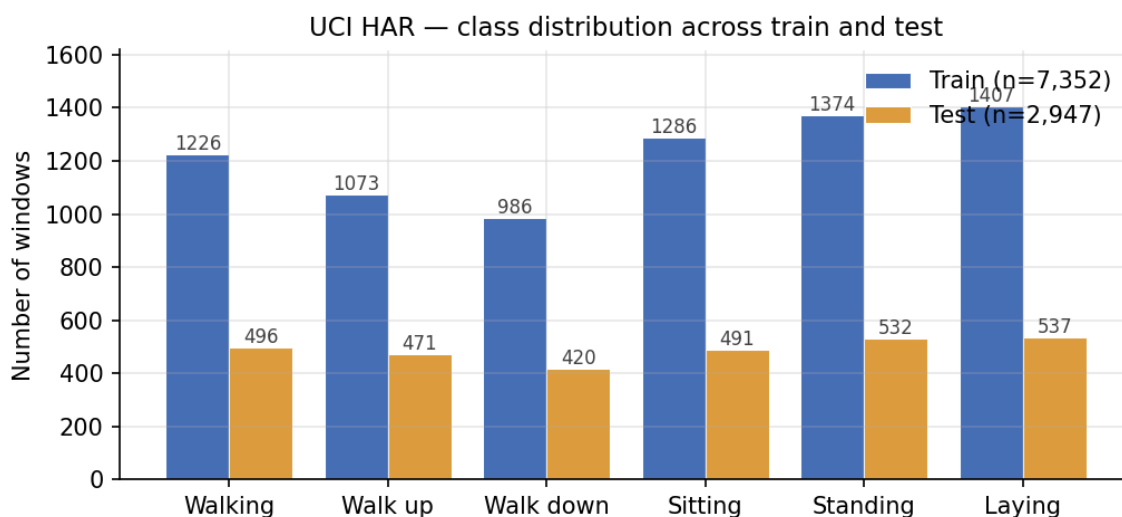


Figure 1 — Class distribution in the UCI HAR training and test sets.

Our first hypothesis, consistent with the physics of the problem, was that static activities (sitting, standing, lying) and dynamic activities (the three walking variants) should be easy to separate, and that the harder problem is within-group separation. Figure 2 shows one example 2.56 s window for each of the six activities. The dynamic activities exhibit clear oscillatory patterns on all three axes, while the static activities are essentially flat — the only useful information they carry is the gravity vector's distribution across axes, which reflects the phone's orientation. When we computed the per-axis means of total acceleration for WALKING windows in the training set we got $x \approx 0.996$ g (≈ 1 g, gravity along the device's x-axis), $y \approx -0.181$ g, $z \approx -0.070$ g, with within-window standard deviations of 0.20–0.23 g on all three axes — confirming that the variance, not the mean, is the discriminating signal.

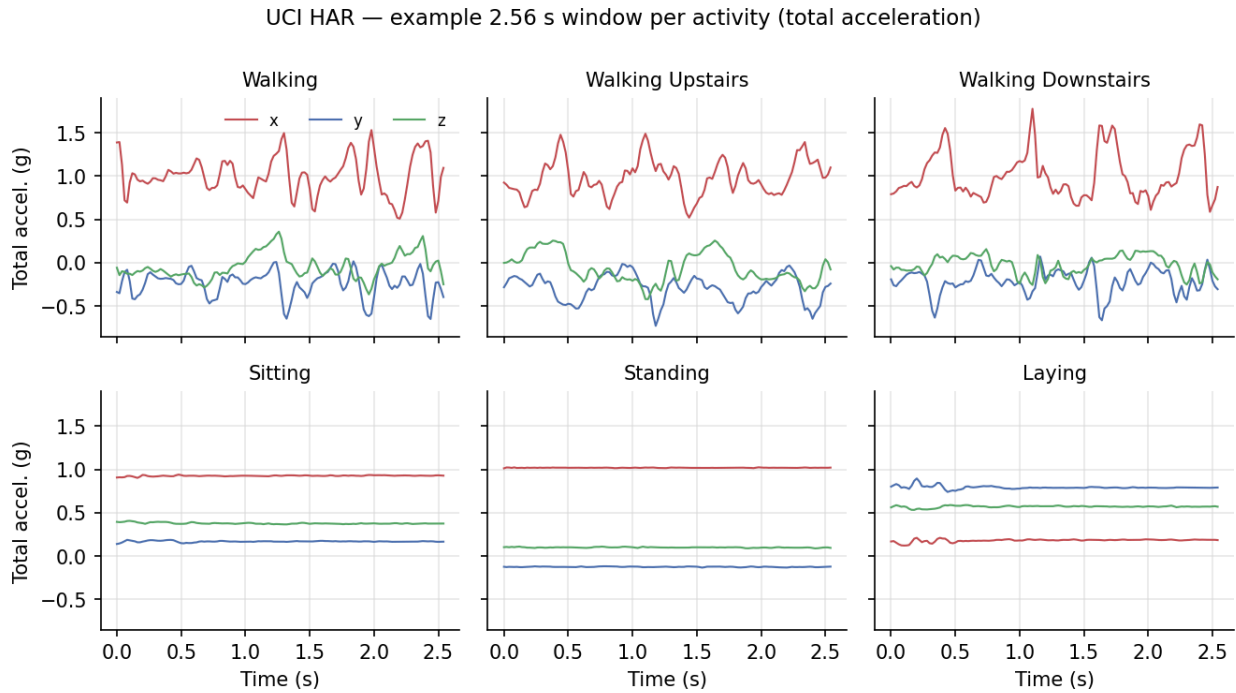


Figure 2 — Example 2.56 s windows of total acceleration (red = x, blue = y, green = z), one per activity. Dynamic activities (top row) show oscillatory patterns; static activities (bottom row) are nearly flat with axis-dependent gravity offsets that reveal the phone's orientation.

We then moved to the frequency domain. Figure 3 shows the average magnitude spectrum of the body-acceleration signal (gravity-removed Euclidean norm) for walking, standing and laying, computed over all training windows of each class. Walking exhibits a dominant peak near 1.6 Hz — exactly the cadence of normal human gait — and rich harmonic content up to about 6 Hz. The two static activities have most of their energy concentrated below 0.5 Hz, indicating near-DC signals with no periodic motion. This is direct empirical justification for the Fourier-based features we committed to in HW1: time-domain statistics alone may struggle to distinguish two static postures with different orientations, but the frequency content provides a clean separation between dynamic and static classes.

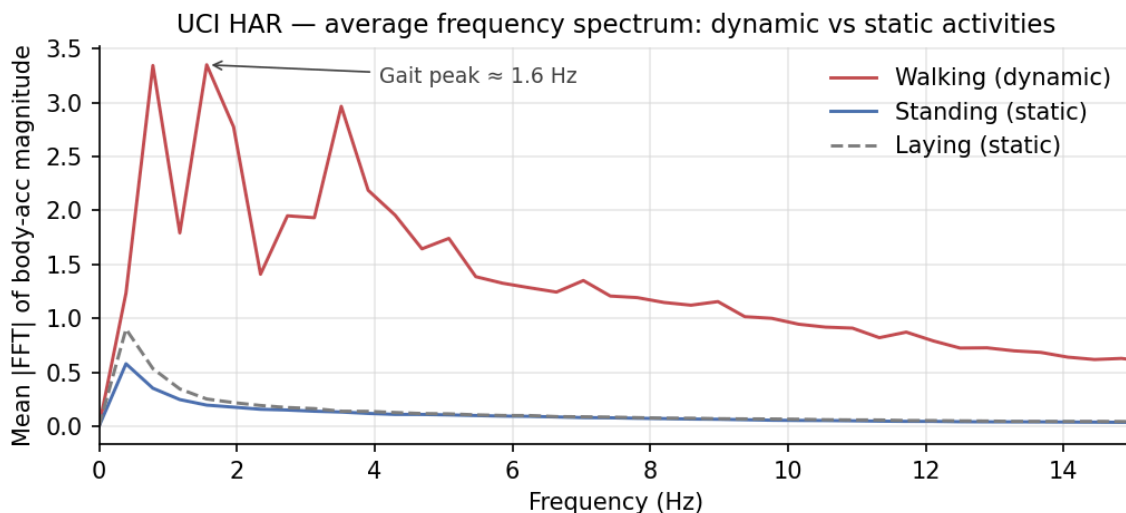


Figure 3 — Mean FFT magnitude of body-acceleration magnitude across all training windows of each class. The walking class shows the characteristic gait-frequency peak near 1.6 Hz; static activities have nearly all their energy below 0.5 Hz.

When we explored our Phyphox recordings, we noticed immediately that the phone's orientation differs substantially from the UCI setup. In our walking-in-hand recording the gravity component sits mostly on the z-axis (≈ 0.91 g) rather than the x-axis as in UCI (≈ 1.0 g). This isn't a data error — it's the natural consequence of holding the phone vertically in the hand instead of mounting it flat on the waist — but it means a model trained on UCI axes without orientation-invariant features will probably get confused. Our planned mitigation is to lean on signal magnitude (the Euclidean norm) and within-window statistics rather than per-axis means. The walking-in-pocket recording is the closest physical match to the UCI waist position, so we expect it to be our most informative demo case.

First findings, summarized: (1) UCI is clean, balanced and well-documented; (2) our time-domain plots confirm a clean dynamic-vs-static separation, while the FFT plot highlights the gait-frequency signature that should drive walking classification; (3) our Phyphox data is internally consistent but differs from UCI in three structural ways — sampling rate, units, and per-axis orientation — all of which we handle in data preparation; (4) the "near real-time" inference claim we made in HW1 is feasible given the 1.28 s stride between windows.

2.4 Data Quality Report

UCI HAR Dataset

We started by checking the integrity of the UCI matrices directly. When we ran `df.isna().sum().sum()` on both `X_train` and `X_test` the result was 0 — there are no missing values anywhere in the raw UCI data. Every feature column stays within the documented `[-1.0, 1.0]` range, the labels are integers 1–6 with no out-of-range entries, and the subject sets for train and test are completely disjoint. This is a remarkably clean dataset, and we treat it as such throughout the project.

- Completeness: 0 missing values in either `X_train` or `X_test` (verified).
- Range: all 561 feature columns are within `[-1.0000, 1.0000]` in both train and test, consistent with the documented normalization.
- Subject leakage: the train/test split is subject-disjoint (we verified the set intersection is empty). This is the correct setup for honest generalization estimation.
- Label correctness: labels are integers 1–6 only; no missing labels.
- Duplicate column names: 42 feature names appear multiple times in `features.txt` (84 columns are affected). The columns themselves contain different data — only the names collide. This is a known dataset quirk and we solve it by appending a suffix at load time (see §3.5).
- Within-window inertial value range: total acceleration $\min \approx -1.25$ g, $\max \approx +1.30$ g — physically plausible for human motion.

Our Phyphox recordings — what we ran into

Self-collecting our own data with Phyphox was a useful reality check, and we hit two issues worth documenting honestly.

Issue 1 — phone orientation changed the per-axis gravity distribution

When we recorded walking with the phone in our front pocket, the orientation of the device relative to the body was different from UCI's waist-mounted Samsung Galaxy S II. The gravity component shifted between axes: in UCI's walking data the dominant gravity component is on the x-axis (≈ 1 g), while in our walking-in-hand recording it sits on the z-axis (≈ 0.91 g), and the walking-in-pocket recording shows yet another distribution. The signals themselves are physically correct; it's the same gravity vector, just projected onto a differently oriented sensor. But for a classifier trained on UCI per-axis features, this is a real distribution shift. We have flagged this in our pipeline and we plan to mitigate it by preferring orientation-invariant features (signal magnitude, energy, frequency content) for the live demo, instead of relying on raw per-axis means.

Issue 2 — missing sensors in some recordings

The raw values themselves contain no NaNs — we verified this for every one of the nine workbooks. The real gap is structural: not every activity has both sensors recorded. For example, walking-in-pocket has a

gyroscope file but no accelerometer file. We handle this explicitly in code: when we assemble each window into the (128, 9) channel layout that matches UCI, we mark the missing channels as NaN and tag the file with a flag. At inference time, the demo loop proactively skips any model that requires the missing channels (e.g. an accelerometer-only classifier will skip walking-in-pocket by design). This way the NaN-marking acts as an explicit "no data" indicator rather than a silent zero that would corrupt predictions.

Other Phyphox observations

- Completeness of recorded values: 0 NaNs across all nine files (the issue above is missing files, not missing samples).
- Sampling stability: time steps are consistent within each file; the measured rate stays within ± 0.4 Hz of 500 Hz across all recordings, so we trust a uniform-rate resampling assumption.
- Cosmetic issues we noticed: the filename for sitting is lowercase "sitting" while the others are title-case; one filename contains a typo ("Poket" instead of "Pocket") and one metadata sheet column is spelled "proeprty" instead of "property". None of these affect the data values.
- Missing classes: our live recordings do not include WALKING_UPSTAIRS or WALKING_DOWNSTAIRS, so the demo can only exercise four of the six UCI classes. This is an experimental scope choice, not a defect — but we record it here because it limits what the live demo can show.
- Single subject: all nine recordings are from one team member on one device. That's enough for a qualitative demo but not for computing generalization metrics, which is exactly why we use Phyphox as a demo and not as a second test set.

What we plan to do about each issue

- Duplicate UCI column names → we append `__dup{k}` to repeated names at load time.
- Phyphox/UCI sampling rate mismatch → we resample 500 Hz → 50 Hz with an antialiasing filter (see §3.3).
- Phyphox/UCI accelerometer unit mismatch → we divide m/s^2 by 9.80665.
- Phyphox orientation differs from UCI → we prefer orientation-invariant features at inference time.
- Phyphox accelerometer and gyroscope live in separate files with slightly different timing → we align them on a common time base when both are needed; otherwise we treat each sensor independently.
- Missing sensors per activity → we fill the missing channels with NaN and flag the file; the demo loop skips models that require those channels.
- Missing stairs activities in Phyphox → we report demo results for the four available activities only; we do not claim generalization to stairs from live data.

Data Preparation

3.1 Rationale for Inclusion / Exclusion

What we include

- UCI HAR engineered feature vectors `X_train.txt` and `X_test.txt` (561 columns each). These are the input to our baseline classifiers (logistic regression, random forest, SVM) — the primary modelling track.
- UCI HAR raw inertial signals (9 channels × 128 samples per window). We need them for the transformer experiment and for like-for-like comparisons with Phyphox data (Phyphox gives us only raw signal).
- UCI activity labels and subject IDs (full coverage). Target variable and grouping variable for subject-aware evaluation.
- All nine of our Phyphox recordings (full coverage). Even files for activities not used in the live demo (walking-with-phone-in-hand) are useful for documenting the orientation gap and for sanity-checking the preprocessing pipeline.

What we exclude

- Phyphox "Absolute" magnitude column. It's the Euclidean norm of (x, y, z), so it would be a perfectly collinear feature. We recompute magnitude when we need it.
- Phyphox device metadata sheet. Not relevant per-window; we recorded the device info once in §2.1.
- UCI `total_acc` channels for the transformer experiment. `body_acc` = `total_acc` – gravity already removes the constant gravity component, which is the more useful input for a motion classifier; keeping both would double dimensionality with redundant information.

3.2 Data Cleaning Report

The UCI dataset arrived clean: no missing values, no out-of-range features, no label errors, no subject leakage. The only cleaning action we had to take was to disambiguate the 84 columns whose names collide in features.txt. We append a numeric suffix (`__dup1`, `__dup2`, ...) to the second and later occurrences of each duplicated name at load time, so we can address the 561 columns unambiguously throughout the project.

Our Phyphox recordings needed more work, but all of it was unit and rate alignment rather than error correction. The raw values were physically plausible and complete in every file. Concretely, we:

- Converted accelerometer values from m/s^2 to g (divide by 9.80665) so the unit matches UCI's accelerometer channels.
- Downsampled from ~500 Hz to 50 Hz using `scipy.signal.resample_poly(up=1, down=10)`, which applies an antialiasing FIR filter before decimation (the rationale is in §3.3). We verified on the walking-in-hand recording that the per-axis means are preserved (z-axis 0.9100 g before, 0.9099 g after), confirming the resampling did not introduce a DC bias.
- Dropped the Absolute magnitude column (redundant).
- Standardised column names across all nine files so they load with a consistent schema (`time_s`, `x`, `y`, `z`).

We deliberately did not apply outlier removal at this stage. Sensor outliers in HAR are often the actual signal — heel-strike spikes during walking, for example — so removing them would destroy class-discriminating information. The Butterworth low-pass filter at 20 Hz that the UCI authors applied (and that our antialiasing filter in the resample step replicates in spirit) is the appropriate noise-control mechanism.

3.3 Construct Data — Derived Attributes

For our baseline modelling track on UCI data, we don't construct any new derived attributes beyond the 561 features the dataset authors already provide. The feature vector already contains the time-domain statistics (mean, std, mad, max, min, sma, energy, iqr, entropy, arCoeff, correlation) and frequency-domain statistics (maxInds, meanFreq, skewness, kurtosis, bandsEnergy) that we committed to in HW1.

For our Phyphox data we have to compute the same family of features from scratch, because Phyphox gives us only the raw signal.

Why we use `resample_poly`, and why this preserves the gait peak

Our first derived step on the Phyphox data is changing the sampling rate. Phyphox samples at ~500 Hz, UCI at 50 Hz — a ratio of exactly 10:1. We intentionally chose `scipy.signal.resample_poly(up=1, down=10)` rather than a naive decimation ("keep every 10th sample") because naive decimation doesn't apply an antialiasing filter and would fold higher-frequency content back into the band we care about. `resample_poly` applies a polyphase FIR filter with the appropriate cutoff (around the new Nyquist, 25 Hz) before downsampling, so the resulting 50 Hz signal is a clean low-pass version of the original.

This matters in practice because the activity-discriminating energy in our data lives well below 25 Hz. The gait peak we measured in the UCI training walking class is at about 1.6 Hz (Figure 3), and the harmonic content extends only to about 6 Hz — far below the new Nyquist limit. We checked empirically that the per-axis means are preserved through the resample (z-axis 0.9100 g before, 0.9099 g after), and visually that the gait oscillations remain intact. We get a 10× reduction in sample count, exact rate compatibility with UCI's windows, and no loss of the frequencies that actually drive classification — the right trade-off for our task.

Features we derive from each Phyphox window

After downsampling to 50 Hz we segment into 128-sample windows with 50% overlap, and for each window we compute, per axis:

- Time-domain: mean, standard deviation, min, max, signal energy ($\sum x^2/N$), interquartile range, signal magnitude area.
- Frequency-domain (from a 128-point FFT): dominant frequency, spectral energy, spectral entropy.
- Magnitude channel: we compute the Euclidean norm magnitude = $\sqrt{x^2 + y^2 + z^2}$ and apply the same time- and frequency-domain features on it. This is an orientation-invariant channel that we expect to transfer better from UCI to our Phyphox demo than per-axis features (because of the orientation issue we documented in §2.4).

For our transformer track we don't need scalar derived attributes — the model consumes the 128×9 raw window directly. We do apply a per-channel z-score normalization, with mean and std computed on the training set only, so the network sees inputs of comparable scale across all 9 channels.

3.3 Construct Data — Generated Records

We do not generate any fully synthetic records. The UCI training set (7,352 windows) is adequate for both our baseline models and a small transformer, and we prefer to report honest performance on the data as delivered rather than introduce augmentation that could hide real generalization failures.

We do however use two restricted forms of "generated records":

- Windowing of our Phyphox recordings: from each ~2-minute stream we generate approximately 92 windows of 128 samples with 50% overlap after downsampling to 50 Hz. This is not synthetic data — every window is a real slice of recorded sensor data — but it is constructive in the CRISP-DM sense, because each window becomes one new record fed to the classifier.
- Per-class prototype windows for visualization only: for each of the six UCI activities we compute the element-wise median across all training windows of that class, yielding one prototype window per class. We use these for exploration plots in the final report; they never reach a model.

3.4 Merged Data

We do not physically merge UCI and Phyphox into a single training table

This is deliberate, and it's the most important point in this section. Mixing our Phyphox recordings into the UCI training set would defeat the purpose of the live demo: we want Phyphox to act as a completely external, previously unseen check on whether the model generalizes from UCI's 30 lab subjects to a new user on a new device. If Phyphox data leaked into training, any "good" result on the live demo would be uninformative.

What we do instead is store both datasets in an identical (128, 9) channel structure (128 samples per window, 9 channels in a fixed order). The shape is shared so that the same trained model can consume either source without code changes, but the data themselves stay in separate arrays with separate roles: UCI is used for training and held-out evaluation, Phyphox is used only for the live demo.

Integration within the UCI HAR Dataset

We assemble the UCI training table by joining three files row-by-row: `X_train.txt` (561 features), `y_train.txt` (activity label) and `subject_train.txt` (subject ID). We perform the same join on the test set. The resulting tables have 7,352 and 2,947 rows with 563 columns each (561 features + activity + subject). The 9 raw inertial channels are kept as a separate 3-D array of shape (n_windows, 128, 9), indexed by the same row position, so a single integer index returns both the engineered feature row and the corresponding raw window.

Integration within our Phyphox recordings

Within Phyphox itself, the accelerometer and gyroscope are recorded in separate workbooks with independent start times. For activities where we have both sensors (sitting, standing, lying, walking-in-hand), we merge the two streams onto a common time base by: (1) computing relative time from each file's Time column; (2) trimming both streams to the shorter recording's duration; (3) resampling both to 50 Hz on the same time grid; (4) concatenating column-wise to produce a 6-channel matrix. For walking-in-pocket we only have the gyroscope file, so we fill the accelerometer channels with NaN and tag the file — at demo time the loop skips any model that needs the missing channels (see §2.4 Issue 2).

Aggregate values

We compute per-subject aggregates (mean confidence per class, fraction of windows predicted correctly) at evaluation time to detect subject-specific failure modes. We do not store these as separate records — we derive them from prediction outputs when we need them.

3.5 Reformatted Data

We apply the following formatting transformations so that our data matches the input contract of the modelling tools (scikit-learn for baselines, PyTorch for the transformer). None of these change the semantic content of the data.

UCI engineered features

- We disambiguate duplicate column names by appending a `__dup{k}` suffix to the second and later occurrences of each repeated name in `features.txt`. Result: 561 unique column names.
- We cast the activity column to a pandas Categorical with explicit order [WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING] for consistent confusion-matrix ordering.
- We use dtype float32 for the feature matrix to reduce memory footprint for the transformer experiment.

UCI raw inertial signals

- We reshape into a single array of shape `(n_windows, 128, 9)` with channels in the fixed order [body_acc_x, body_acc_y, body_acc_z, body_gyro_x, body_gyro_y, body_gyro_z, total_acc_x, total_acc_y, total_acc_z]. We record the channel order explicitly to avoid silent permutations downstream.
- We compute per-channel mean and standard deviation on the training set only, and apply z-score normalization with the same statistics to both train and test windows.

Our Phyphox recordings

- Convert accelerometer values from m/s^2 to g (divide by 9.80665).
- Resample from ~ 500 Hz to exactly 50 Hz with `resample_poly(up=1, down=10)`.
- Segment each recording into 128-sample windows with 64-sample stride (50% overlap), matching the UCI windowing exactly.
- Assemble each window into the same (128, 9) channel layout used for UCI raw signals where possible. Where a sensor is missing for the activity (e.g. accelerometer for walking-in-pocket), we fill the missing channels with NaN and flag the file; the demo loop skips models that require the missing channels.
- Save the prepared train/test arrays and the prepared Phyphox windows as NumPy .npz files with a small JSON sidecar describing the channel order, units and sampling rate. We can reuse the preprocessed data across notebooks without re-running the full pipeline.

After all formatting steps, the inputs to our modelling phase are: a $(7,352 \times 561)$ feature matrix and a $(2,947 \times 561)$ feature matrix for the baseline models; a $(7,352 \times 128 \times 9)$ array and a $(2,947 \times 128 \times 9)$ array for the transformer; and a small set of $(\sim 92 \times 128 \times 6)$ arrays per Phyphox recording for the live demo. We carry labels and subject IDs alongside each array in 1-D companion arrays of matching length.